DEPARTMENT OF COMPUTER SCIENCE
RAJAGIRI COLLEGE OF SOCIAL SCIENCES
(Autonomous)
KALAMASSERY - KOCHI - 683104



MASTER OF COMPUTER APPLICATIONS

DBMS LAB RECORD

NAME : MERIN JAI

SEMESTER : _____

REGISTER NO. : _____

DEPARTMENT OF COMPUTER SCIENCE
RAJAGIRI COLLEGE OF SOCIAL SCIENCES
(Autonomous)
KALAMASSERY - KOCHI - 683104



MASTER OF COMPUTER APPLICATIONS

DBMS
LAB RECORD

NAME : MERIN JAI

SEMESTER : _____

REGISTER NO. : _____



DEPARTMENT OF COMPUTER SCIENCE
RAJAGIRI COLLEGE OF SOCIAL SCIENCES
(Autonomous)
KALAMASSERY - KOCHI – 683104

MASTER OF COMPUTER APPLICATIONS

CERTIFICATE

NAME : MERIN JAI

SEMESTER : _____

REGISTER NO. : _____

Certified that this is a bonafide record of work done by the student in the Software Laboratory of Rajagiri Department of Computer Science, Kalamassery.

Faculty in Charge

Dean, Computer Science

Internal Examiner

External Examiner

Place : Kalamassery
Date :

Table of Contents

Activity **Page No**

1. **E-R Diagram & Table Design**

2. **Practice SQL Data Definition Language(DDL) commands**
   - 2.1 Table creation and alteration

3. **Practice SQL Data Manipulation Language (DML) commands**
   - 3.1 Row insertion, deletion and updating
   - 3.2 Retrieval of data (Simple select query and select with where options (include all relational and logical operators)
   - 3.3 Functions: Numeric Data, Character Conversion and
   - 3.4 Group functions
   - 3.5 Data manipulations using date functions
   - 3.6 Set Operations
   - 3.7 Illustration of Group by Having Clause
   - 3.8 Sub Queries
   - 3.9 Retrieving from multiple tables (Illustrate with Join Clause also)

4. **SQL Views**
5. **Practice PL/SQL**
   - 5.1 Introductory programs
   - 5.2 Illustration of Cursors
   - 5.3 Illustration of Procedures
   - 5.4 Illustration of functions
   - 5.5 Illustration of Triggers

Activity # 1
ER Diagram & Table Design

Description :

- E-R Diagram and table reduction
- Table descriptions

Date :

ER Diagram & Table Design

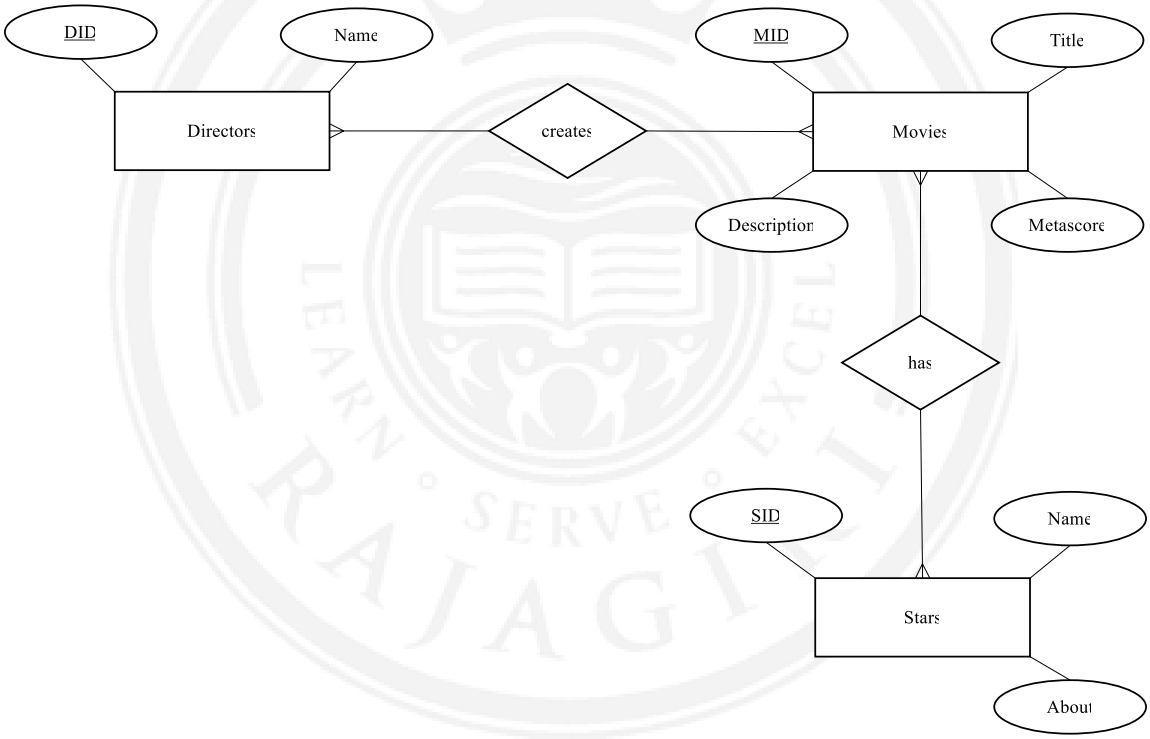


Table name: Directors
Description: Used to store Directors Information

Attribute	Data Type	Constraints
DId	Int	Primary Key/ Not Null
Name	Varchar2(40)	Not Null

Table name: Stars
Description: Used to store Stars Information

Attribute	Data Type	Constraints
SId	Int	Primary Key/ Not Null
Name	Varchar2(40)	Unique
About	Varchar2(100)	

Table name: Movies
Description: Used to store Movies Information

Attribute	Data Type	Constraints
MId	Int	Primary Key/ Not Null
Title	Varchar2(40)	Not Null
R_date	Date	
Image_url	Varchar2(100)	
Certificate	Varchar2(20)	
Runtime	Number(3,2)	
ImdbRating	Number (3,1)	By default 0
Description	Text(100)	By default Null
Metascore	Number (3,1)	By default 0
Votes	Int	By default 0
Gross	Number(10,2)	Gross amount should be greater than 10000

Table name: MoviesDirectors
Description: Used to store Movie Directors Information

Attribute	Data Type	Constraints	
MoviesId	Int	Foreign Key references Id of **Movies** table	
DirectorsId	Int	Foreign Key references Id of **Directors** table	Primary Key

Table name: MoviesStars
Description: Used to store Movie Stars Information

Attribute	Data Type	Constraints	
MoviesId	Int	Foreign Key references Id of **Movies** table	
StarsId	Int	Foreign Key references Id of **Stars** table	Primary Key

Activity # 2
Practice SQL Data Definition Language(DDL) commands

Description : 2.1 Table creation and alteration

Date :

Query

1. Create the tables based on the above description.

```
SQL> create table Directors(Did int primary key, name varchar2(40) not null);

Table created.
```

```
SQL> create table stars(sid int primary key, name varchar2(40) unique, about varchar2(100));

Table created.

SQL> desc stars;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SID                                       NOT NULL NUMBER(38)
 NAME                                               VARCHAR2(40)
 ABOUT                                              VARCHAR2(100)
```

```
SQL> create table movies(mid int primary key, title varchar2(40) not null, R_date date, Image_url varcha
r2(100), Certificate varchar2(20), Runtime Number(3,2), ImdbRating Number(3,1) Default 0, Description va
rchar2(100) Default NULL,Metascore Number(3,1) Default 0,Votes int Default(0), Gross Number(10,2) CHECK(
gross>10000));

Table created.

SQL> desc movies
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 R_DATE                                             DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 GROSS                                              NUMBER(10,2)
```

```
SQL> create table MoviesDirectors(mid int references movies(mid), did int references directors(did),prim
ary key(mid,did));

Table created.

SQL> desc MoviesDirectors;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 DID                                       NOT NULL NUMBER(38)
```

```
SQL> create table MoviesStars(mid int references movies(mid), sid int references stars(sid),primary key(
mid,sid));

Table created.

SQL> desc MoviesStars;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 SID                                       NOT NULL NUMBER(38)
```

2. Add a column 'DOB' to Stars table.

```
SQL> alter table stars add dob date;

Table altered.

SQL> desc stars;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 SID                                       NOT NULL NUMBER(38)
 NAME                                               VARCHAR2(40)
 ABOUT                                              VARCHAR2(100)
 DOB                                                DATE
```

3. Drop the column 'Gross' in Movies table.

```
SQL> alter table Movies drop column gross;

Table altered.

SQL> desc Movies;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 R_DATE                                             DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
```

4. Add column 'Language' in Movies table.

```
SQL> alter table movies add language varchar2(20);

Table altered.

SQL> desc movies;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 LANGUAGE                                           VARCHAR2(20)
```

5. Add column Gross Number(10,2) in Movies table

```
SQL> alter table movies add Gross number(10,2);

Table altered.

SQL> desc movies
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 LANGUAGE                                           VARCHAR2(20)
 HIT                                                NUMBER(1)
 ENTRY_DATE                                         DATE
 GROSS                                              NUMBER(10,2)
```

6. Change the name of the column 'R_date' in Movies table to Releasedate.
   Releasedate.

```
SQL> alter table movies rename column R_date to Releasedate;

Table altered.

SQL> desc movies
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 LANGUAGE                                           VARCHAR2(20)
 HIT                                                NUMBER(1)
 ENTRY_DATE                                         DATE
 GROSS                                              NUMBER(10,2)
```

7. Add a column 'Age' in Directors table as Number. Age must be 7 years or above.

```
SQL> alter table directors add age Number(10) check(age>=7);

Table altered.

SQL> desc directors
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DID                                       NOT NULL NUMBER(38)
 NAME                                      NOT NULL VARCHAR2(30)
 AGE                                                NUMBER(10)
```

8. Add a new column ‘Hit’ in Movies table with datatype Number(1) and by default 0.

```
SQL> alter table Movies add hit Number(1) default 0;

Table altered.

SQL> desc movies
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 LANGUAGE                                           VARCHAR2(20)
 HIT                                                NUMBER(1)
```

9. Add a new column ‘Entry_date’ in Movies table to record the date on which the movie details are entered in the data base.

```
SQL> alter table movies add entrydate date;

Table altered.

SQL> desc movies
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 IMAGE_URL                                          VARCHAR2(100)
 CERTIFICATE                                        VARCHAR2(20)
 RUNTIME                                            NUMBER(3,2)
 IMDBRATING                                         NUMBER(3,1)
 DESCRIPTION                                        VARCHAR2(100)
 METASCORE                                          NUMBER(3,1)
 VOTES                                              NUMBER(38)
 LANGUAGE                                           VARCHAR2(20)
 HIT                                                NUMBER(1)
 ENTRY_DATE                                         DATE
 GROSS                                              NUMBER(10,2)
 ENTRYDATE                                          DATE
```

10. Destroy the table MoviesStars and recreate it.

```
SQL> drop table Moviesstars;

Table dropped.

SQL> create table MoviesStars(mid int references movies(mid), sid int references stars(sid),primary key(
mid,sid));

Table created.
```

11. Change the size of the Director's name to 30.

```
SQL> alter table directors modify name varchar2(30);

Table altered.

SQL> desc directors;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 DID                                       NOT NULL NUMBER(38)
 NAME                                      NOT NULL VARCHAR2(30)
 AGE                                                NUMBER(10)
```

12. Add the following check constraints:
    - Releasedate should be less than the Entry_date in the Movies table.

```
SQL> alter table movies add check(releasedate<entry_date);

Table altered.
```

    - Language of movies should be Malayalam, English, Tamil or Hindi.

```
SQL> alter table movies add check(language in('Malayalam','English','Hindi','Tamil'));

Table altered.
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.1 Row insertion, deletion and updating

Date :

Query

1. Insert the appropriate data (10 rows) for the tables with respect to defined datatypes, size and constraints.

   Movies table

```
SQL> insert into movies values(101,'Inception','16-Jul-2010','http://www.movieimage.com/_Inception','PG-13',2.47,8.8,'Mind-bending thriller',74,2046235,'English',1,'01-Jan-2011',12345678.12);

1 row created.

SQL> insert into movies values(102,'Mayaanadhi','22-Dec-2022','http://www.movieimage.com/mayanadhi','U/A',2.27,8,'Romantic Drama',75,20900,'Malayalam',1,'01-Jan-2023',52345678);

1 row created.

SQL> insert into movies values(103,'Oppam','08-Sep-2016','http://www.movieimage.com/oppam','U',2.62,7.5,'Crime Thriller',70,10545,'Malayalam',0,'01-Dec-2016',1500000);

1 row created.

SQL> insert into movies values(104,'Avatar','18-Dec-2020','http://www.movieimage.com/_avatar','PG-13',2.7,7.8,'Epic sci-fi adventure',83,1181175,'English',1,'01-Sep-2021',211787.97);

1 row created.

SQL> insert into movies values(105,'Daddy Cool','14-Aug-2009','http://example.com/daddy_cool.jpg','U',3,5.8,'A quirky family comedy with a touch of mystery',5.6,4200,'Malayalam',1,'07-Sep-2009',5330000);

1 row created.

SQL> insert into movies values(106,'The Avengers','04-May-2012','http://www.movieimage.com/_avengers','PG-13',2.38,8,'Superhero ensemble',69,1353551,'English',0,'01-Sep-2013',621518.81);

1 row created.

SQL> insert into movies values(107,'Pulp Fiction','14-Oct-1994','http://www.movieimage.com/_4','R',2.57,8.9,'Quirky crime film',94,1900251,'English',1,'01-Sep-1995',5111213.93);

1 row created.

SQL> insert into movies values(109,'Lucifer','28-Mar-2019','http://example.com/lucifer.jpg','PG-13',2.5,8.5,'A political thriller with unexpected twists',8.2,32000,'Malayalam',1,'30-Mar-2019',515000);

1 row created.

SQL> insert into movies values(110,'Forrest Gump','06-Jul-1994','http://www.movieimage.com/_6','PG-13',2.37,8.8,'Heartwarming drama',82,2176822,'English',0,'01-Sep-1995',67811.18);

1 row created.

SQL> insert into movies values(108,'Master','23-Jan-2023','http://example.com/movie2.jpg','R',2,6.9,'A suspenseful thriller that takes place under the cover of darkness.',7.5,9872,'Hindi',0,'29-Aug-2023',31500.5
);

1 row created.
```

Directors Table

```
SQL> insert into directors values(115,'Priyadarshan',64);

1 row created.

SQL> insert into directors values(116,'Lijo Jose Pellissery',42);

1 row created.

SQL> insert into directors values(117,'Aashiq Abu',42);

1 row created.

SQL> insert into directors values(118,'Christopher Nolan',51);

1 row created.

SQL> insert into directors values(119,'Steven Spielberg',75);

1 row created.

SQL> insert into directors values(126,'Prithviraj Sukumaran',42);

1 row created.

SQL> insert into directors values(121,'Martin Scorsese',79);

1 row created.

SQL> insert into directors values(122,'Ridley Scott',83);

1 row created.

SQL> insert into directors values(125,'Lokesh Kanagaraj',35);

1 row created.

SQL> insert into directors values(124,'Robert Zemeckis',70);
```

Stars table

```
SQL> insert into stars values(301,'Mohanlal','Legendary Malayalam actor','21-May-1960');

1 row created.

SQL> insert into stars values(302,'Zoe Saldana','Hollywood actor','05-May-1964');

1 row created.

SQL> insert into stars values(303,'Prithviraj','Versatile actor','16-Oct-1982');

1 row created.

SQL> insert into stars values(304,'Tovino Thomas','Acclaimed actor','08-Aug-1982');

1 row created.

SQL> insert into stars values(305,'Leonardo DiCaprio','Award-winning actor','11-Nov-1974');

1 row created.

SQL> insert into stars values(306,'Robert Downey Jr.','Iron Man','04-Apr-1965');

1 row created.

SQL> insert into stars values(307,'Vijay','Popular Tamil actor','22-Jun-1974');

1 row created.

SQL> insert into stars values(308,'Emma Watson','Hermione Granger','15-Apr-1990');

1 row created.
```

```
SQL> insert into stars values(310,'Mammootty','Iconic Malayalam actor','07-Sep-1951');

1 row created.

SQL> insert into stars values(309,'Sreenath Bhasi','Malayalam actor','04-May-1985');

1 row created.
```

MoviesDirectors Table

```
SQL> insert into moviesdirectors values(101,118);

1 row created.

SQL> insert into moviesdirectors values(102,117);

1 row created.

SQL> insert into moviesdirectors values(103,115);

1 row created.

SQL> insert into moviesdirectors values(104,118);

1 row created.

SQL> insert into moviesdirectors values(105,117);

1 row created.

SQL> insert into moviesdirectors values(106,124);

1 row created.

SQL> insert into moviesdirectors values(107,122);

1 row created.

SQL> insert into moviesdirectors values(108,125);

1 row created.

SQL> insert into moviesdirectors values(109,126);

1 row created.

SQL> insert into moviesdirectors values(110,121);

1 row created.
```

Moviesstars Table

```
SQL> insert into moviesstars values(101,305);

1 row created.

SQL> insert into moviesstars values(102,304);

1 row created.

SQL> insert into moviesstars values(103,301);

1 row created.

SQL> insert into moviesstars values(104,302);

1 row created.

SQL> insert into moviesstars values(105,310);

1 row created.

SQL> insert into moviesstars values(106,306);

1 row created.

SQL> insert into moviesstars values(107,305);

1 row created.

SQL> insert into moviesstars values(108,307);

1 row created.

SQL> insert into moviesstars values(109,303);

1 row created.

SQL> insert into moviesstars values(109,301);

1 row created.
```

2. Change value of Hit to 1 where 'Votes' greater than or equal to 90.

```
SQL> update movies set hit=1 where votes>=90;

6 rows updated.
```

3. Create table **IndustryHit** with the following columns:
   Id
   Title
   Releasedate
   Language
   Votes
   Gross

   The data types and null characteristics for these columns should be the same as the corresponding columns in the **Movies** table described at the beginning of the lab exercise.

```
SQL> create table IndustryHit(mid int primary key, title varchar2(40) not null, ReleaseDate date, Langua
ge varchar2(20),Votes Number(38),Gross int);

Table created.

SQL> desc IndustryHit;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 MID                                       NOT NULL NUMBER(38)
 TITLE                                     NOT NULL VARCHAR2(40)
 RELEASEDATE                                        DATE
 LANGUAGE                                           VARCHAR2(20)
 VOTES                                              NUMBER(38)
 GROSS                                              NUMBER(38)

SQL>
```

4. New movies hit the box office; their data is as follows:
   - Id: 1014, 1021, 1032
   - Title: 2018: Everyone is a Hero, Oppenheimer, Maamannan
   - Releasedate: 5 May 2023, 21 July 2023, 29 June 2023
   - Language: Malayalam, English, Tamil
   - Votes: 97, 96, 95
   - Gross: 750000000, 500000000, 505000000
   - Add the new employees to the **IndustryHit** table.

```
SQL> Insert into IndustryHit values(1014,'Everyone is a hero','5-may-2023','Malayalam',97,750000000);

1 row created.

SQL> Insert into IndustryHit values(1021,'Oppenheimer','21-Jul-2023','English',96,500000000);

1 row created.

SQL> Insert into IndustryHit values(1032,'Maamannan','29-Jun-2023','Tamil',95,505000000);

1 row created.
```

5. Insert data into the new **IndustryHit** table.

```
SQL> insert into IndustryHit values(1056,'PuliMurugan','27-Oct-2023','Malayalam',98,625000000);

1 row created.

SQL> insert into IndustryHit values(1011,'Gulmohar','25-May-2023','Hindi',95,504010000);

1 row created.

SQL>  insert into IndustryHit values(1051,'Maari','09-Jan-2023','Tamil',97,514350000);

1 row created.

SQL>
```

6. Insert data into the **IndustryHit** table by copying the appropriate columns in the **Movies** table for those Movies that have Votes greater than or equal to 95.

```
SQL> Insert into IndustryHit select mid,title,releasedate,language,votes,gross from movies where votes>=
95;

2 rows created.
```

7. Movie Oppenheimer got a Metascore of 80. Make the appropriate data change.

```
SQL> update movies set metascore=80 where title='Oppenheimer';

1 row updated.
```

8. Delete all movies whose Metascore is less than 50.

```
SQL> delete from movies where metascore<50;
```

9. Movie 'Voice Of Sathyanathan' was released.
    - For 'Voice Of Sathyanathan' enter the following data:
        - Id: 1015
        - Title: Voice Of Sathyanathan
        - Releasedate: 28 July 2023
        - Image_url: https://m.media-amazon.com/imak2M_.jpg
        - Certificate: U
        - Runtime: 2.10
        - ImdbRating: 7.4
        - Description: A man's life becomes increasingly complicated after his neighbor is injured in a dispute over a fence.
        - Metascore: 60
        - Votes: 90
        - Gross: 109500000

```
SQL>  insert into movies values(1015,'Voice Of Sathyanathan','29-Jul-2023','https://m.media-amazon.com/i
mak2M_.jpg','U',2.10,7.4,'A mans life becomes increasingly complicated after his neighbor is injured in
a dispute over a fence',60,90,'01-Sep-2023','Malayalam',1,109500000);

1 row created.
```

10. Delete all rows from **IndustryHit and drop the IndustryHit table.**

```
SQL> delete from industryhit;

8 rows deleted.

SQL> drop table industryhit;

Table dropped.
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.2 Retrieval of data (Simple select query and select with 'where' options (include all relational and logical operators)

Date :

Query

1. List details of all movies

```
SQL> select * from movies;

       MID TITLE                                    RELEASEDA IMAGE_URL
                                  CERTIFICATE              RUNTIME IMDBRATING DESCRIPTION
                                  METASCORE      VOTES ENTRYDATE LANGUAGE                    HIT      GROSS
---------- ---------------------------------------- --------- ----------------------------------------------------------------------------------------------------
------------------------------ -------------------- ---------- ---------- -------------------- ---------- ----------
------------------------------------------------------------------------------------------------------------------------
      1015 Voice Of Sathyanathan                    29-JUL-23 https://m.media-amazon.com/imak2M_.jpg
                                  U                          2.1        7.4 A mans life becomes increasingly complicated after his n
eighbor is injured in a dispute over a fence       60         90 01-SEP-23 Malayalam                  1  109500000
       101 Inception                                16-JAN-20 http://www.movieimage.com/_Inception
                                  PG-13                    2.47        8.8 Mind-bending thriller
                                                         74         95 01-JUN-20 English                    1   12345678
       102 Mayaanadhi                               22-DEC-22 http://www.movieimage.com/mayanadhi
                                  U/A                      1.27        6.4 Romantic Drama
                                                         75         94 01-JAN-23 Malayalam                  1 5234500678
       103 Oppam                                    16-JUN-20 http://www.movieimage.com/oppam
                                  U                        2.62        7.5 Crime Thriller
                                                         90         93 01-DEC-20 Malayalam                  1    1500000
       104 Avatar                                   16-JUN-20 http://www.movieimage.com/_avatar
                                  PG-13                     2.7        5.9 Epic sci-fi adventure
                                                         83         91 01-SEP-21 English                    1     211788
       105 Daddy Cool                               14-FEB-09 http://example.com/daddy_cool.jpg
                                  U                         3        5.8 A quirky family comedy with a touch of mystery
                                                       5.6         89 07-SEP-09 Malayalam                  1  650030000
       106 The Avengers                             04-MAY-12 http://www.movieimage.com/_avengers
                                  PG-13                  2.38          8 Superhero ensemble
                                                         69         90 01-SEP-13 English                    1     621519
       107 Pulp Fiction                             14-FEB-20 http://www.movieimage.com/_4
                                  R                        2.57        8.9
                                                         94         78 01-SEP-20 English                    1    5111214
       108 Master                                   23-JAN-23 http://example.com/movie2.jpg
                                  R                          2        6.9 A suspenseful thriller that takes place under the cover
of darkness.                                         7.5         45 29-AUG-23 Tamil                      0      31501
       109 Lucifer                                  28-MAR-20 http://example.com/lucifer.jpg
                                  PG-13                    2.5        8.5 A political thriller with unexpected twists
                                                       8.2         34 30-MAR-20 Malayalam                  1     515000
       110 Oppenheimer                              21-FEB-23 http://example.com/openhei.jpg
                                  R                        2.75        8.1
                                                         80         96 25-JUL-23 English                    1  505000000

11 rows selected.
```

2. List Title, Votes, ReleaseYear, Gross where Gross collection greater than 5000,000,00. Sequence the results in descending order by Gross.

```
SQL> select title,votes,releasedate,gross from movies where gross>=500000000 order by gross desc;

TITLE                                          VOTES RELEASEDA      GROSS
---------------------------------------- ---------- --------- ----------
Mayaanadhi                                         94 22-DEC-22 5234500678
Daddy Cool                                         89 14-FEB-09  650030000
Oppenheimer                                        96 21-FEB-23  505000000
```

3. Retrieve the titles and years of Tamil movies released in 2022.

```
SQL> select title, releasedate from movies where language='Tamil' and TO_CHAR(releasedate,'YYYY')=2022;

no rows selected
```

4. Get the titles, years, and meta scores of movies sorted in descending order of meta scores.

```
SQL> select title,releasedate,metascore from movies order by metascore desc;

TITLE                                    RELEASEDA  METASCORE
---------------------------------------- --------- ----------
Pulp Fiction                             14-FEB-20         94
Oppam                                    16-JUN-20         90
Avatar                                   16-JUN-20         83
Oppenheimer                              21-FEB-23         80
Mayaanadhi                               22-DEC-22         75
Inception                                16-JAN-20         74
The Avengers                             04-MAY-12         69
Voice Of Sathyanathan                    29-JUL-23         60
Lucifer                                  28-MAR-20        8.2
Master                                   23-JAN-23        7.5
Daddy Cool                               14-FEB-09        5.6

11 rows selected.
```

5. List titles, years, languages, dates and votes of all Malayalam and English movies released before 2022 and ImdbRating less than 7. The list should be ordered by Title.

```
SQL> select title,releasedate,language,entrydate,votes from movies where language in ('Malayalam','English') and releasedate<'01-Jan-2022' and  IMDBRATING<7 order by title;

TITLE                                    RELEASEDA LANGUAGE            ENTRYDATE      VOTES
---------------------------------------- --------- -------------------- --------- ----------
Avatar                                   16-JUN-20 English              01-SEP-21         91
Daddy Cool                               14-FEB-09 Malayalam            07-SEP-09         89
```

6. List all the movies whose title starts with 'Oppen'. Order the result by descending order of their id.

```
SQL> select mid,title from movies where title like 'Oppen%' order by mid desc;

       MID TITLE
---------- ----------------------------------------
       110 Oppenheimer
```

7. List Hit movies released in 2022 and 2023. Order the result by ascending order of their Titles.

```
SQL> select * from movies where hit=1 and to_char(releasedate,'YYYY') in (2022,2023) order by title;

       MID TITLE                                RELEASEDA IMAGE_URL
                                                
```

8. Retrieve movies with a runtime between 1.5 and 2.5 hours.

```
SQL> select * from movies where runtime between 1.5 and 2.5;

      MID TITLE                                RELEASEDA IMAGE_URL                                                
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.3 Functions: Numeric Data, Character Conversion and

Group functions

Date :

Query

1. Illustrate the different numeric functions using dual table (power, round, ceil, floor, abs, exp, greatest, least, mod, trunc, round, sign, sqrt etc.)

```
SQL> select power(5,6) ,round(80.766761,4) , ceil(80.35641), floor(59.4) , abs(6.55) , exp(25) , greatest(503,800,632), least(310,228,126), mod(25,3) , trunc(125.8150,-2) , sign(-55) , sqrt
(789) from dual;

POWER(5,6) ROUND(80.766761,4) CEIL(80.35641) FLOOR(59.4)  ABS(6.55)    EXP(25) GREATEST(503,800,632) LEAST(310,228,126)  MOD(25,3) TRUNC(125.8150,-2)  SIGN(-55)  SQRT(789)
---------- ------------------ --------------- ----------- ---------- ---------- --------------------- ------------------ ---------- ------------------- ---------- ----------
     15625            80.7668              81          59       6.55 7.2005E+10                   800                126          1                 100         -1 28.0891438
```

2. Illustrate the character functions (upper, lower, initcap, length, concat, ascii, substr, ltrim, rtrim, trim, translate, instr, chr,Lpad,Rpadetc) using the table **Movies.**

```
SQL> select upper(title) UPPER,lower(title) as LOWER,initcap(title),length(description) LENGTH,concat(mid,metascore) CONCAT,ascii(certificate) ASCII,substr(title,0,4),ltrim(title,'M') LTRIM
,rtrim(title,'r') as RTRIM,trim(title),translate(title, 'aeiou', 'AEIOU') from movies;

UPPER                                LOWER                                INITCAP(TITLE)                       LENGTH CONCAT
                                   ASCII SUBS LTRIM                             RTRIM                              TRIM(TITLE)                                  TRANSLATE(TITLE,'AE
IOU','AEIOU')
------------------------------------ ---------- ------------------------------------ ------------------------------------ ---------- ------------------------------------ ------------------------------------
------------------------------------ ---------- ---- ------------------------------------ ------------------------------------ ---------- ------------------------------------ ------------------------------------
--------------
VOICE OF SATHYANATHAN                 voice of sathyanathan                Voice Of Sathyanathan                 100 101560
                                    85 Voic Voice Of Sathyanathan              Voice Of Sathyanathan              Voice Of Sathyanathan                        VOIcE Of SAthyAnAth
An
INCEPTION                             inception                            Inception                             21 10174
                                    80 Ince Inception                          Inception                          Inception                                    IncEptIOn
MAYAANADHI                            mayaanadhi                           Mayaanadhi                            14 10275
                                    85 Maya ayaanadhi                          Mayaanadhi                         Mayaanadhi                                   MAyAAnAdhI
OPPAM                                 oppam                                Oppam                                 14 10390
                                    85 Oppa Oppam                              Oppam                              Oppam                                        OppAm
AVATAR                                avatar                               Avatar                                21 10483
                                    80 Avat Avatar                             Avata                              Avatar                                       AvAtAr
DADDY COOL                            daddy cool                           Daddy Cool                            46 1055.6
                                    85 Dadd Daddy Cool                         Daddy Cool                         Daddy Cool                                   DAddy COOl
THE AVENGERS                          the avengers                         The Avengers                          18 10669
                                    80 The  The Avengers                       The Avengers                       The Avengers                                 ThE AvEngErs
PULP FICTION                          pulp fiction                         Pulp Fiction                          10794
                                    82 Pulp Pulp Fiction                       Pulp Fiction                       Pulp Fiction                                 PUlp FIctIOn
MASTER                                master                               Master                                68 1087.5
                                    82 Mast aster                              Maste                              Master                                       MAstEr
LUCIFER                               lucifer                              Lucifer                               43 1098.2
                                    80 Luci Lucifer                            Lucife                             Lucifer                                      LUcIfEr
OPPENHEIMER                           oppenheimer                          Oppenheimer                           11080
                                    82 Oppe Oppenheimer                        Oppenheime                         Oppenheimer                                  OppEnhEImEr

11 rows selected.
```

3. Illustration of conversion functions- to_number, to_char(numberconversion), to_char(dateconversion)

```
SQL> SELECT TO_NUMBER('365.1'),TO_CHAR(9876.54, '9999.99'),to_char(sysdate,'DD-MM-YY') FROM dual;

TO_NUMBER('365.1') TO_CHAR( TO_CHAR(
------------------ -------- --------
             365.1  9876.54 07-09-23
```

4. Count the total no. of Movies

```
SQL> select count(*) AS COUNT from movies;

     COUNT
----------
        11
```

5. Calculate the average votes of movies.

```
SQL> select avg(votes) from movies;

AVG(VOTES)
----------
81.3636364
```

6. Determine the maximum and minimum collection of movies. Rename the output as Max_Coll and Min_Coll respectively.

```
SQL> select max(gross) Max_Coll, min(gross) Min_Coll from movies;

  MAX_COLL   MIN_COLL
---------- ----------
5234500678      31501
```

7. Count the number of movies crossed the collection 50,00,00,000.

```
SQL> select count(*) from movies where gross>500000000;

  COUNT(*)
----------
         3
```

8. Count the hit movies of 2021.

```
SQL> select count(*) from movies where hit=1 and to_char(releasedate,'YYYY')=2021;

  COUNT(*)
----------
         0
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.4 Data manipulations using date functions

Date :

Query

1. Provide a list of all movies which were released on June 16, 2020. Display the year and month of the released date and the Id. Sort the result by Id. Name the derived columns YEAR and MONTH.

```
SQL> select mid,title,to_char(releasedate,'YYYY') YEAR,to_char(releasedate,'MONTH') MONTH from movies where releasedate='16-Jun-2020' order by mid;

      MID TITLE                                    YEAR MONTH
--------- ---------------------------------------- ---- ---------
      103 Oppam                                    2020 JUNE
      104 Avatar                                   2020 JUNE
```

2. List the number of months between release date and entry date of each movie.

```
SQL> select mid,title,round(months_between(ENTRYDATE,RELEASEDATE))MONTHS from movies;

       MID TITLE                                     MONTHS
---------- ---------------------------------------- ----------
      1015 Voice Of Sathyanathan                             1
       101 Inception                                         5
       102 Mayaanadhi                                        0
       103 Oppam                                             6
       104 Avatar                                           15
       105 Daddy Cool                                        7
       106 The Avengers                                     16
       107 Pulp Fiction                                      7
       108 Master                                            7
       109 Lucifer                                           0
       110 Oppenheimer                                       5

11 rows selected.
```

3. List the Entry_date in the format 'DD-Month-YY'.

```
SQL> SELECT mid,title,TO_CHAR(Entrydate, 'DD-Month-YY') AS Entry_Date FROM Movies;

       MID TITLE                                     ENTRY_DATE
---------- ---------------------------------------- ---------------
      1015 Voice Of Sathyanathan                     01-September-23
       101 Inception                                 01-June     -20
       102 Mayaanadhi                                01-January  -23
       103 Oppam                                     01-December -20
       104 Avatar                                    01-September-21
       105 Daddy Cool                                07-September-09
       106 The Avengers                              01-September-13
       107 Pulp Fiction                              01-September-20
       108 Master                                    29-August   -23
       109 Lucifer                                   30-March    -20
       110 Oppenheimer                               25-July     -23

11 rows selected.
```

4. List the date, 8 days after today's date.

```
SQL> SELECT SYSDATE + 8 AS Eight_Days_After FROM DUAL;

EIGHT_DAY
---------
12-SEP-23
```

5. List all the movies which were released in the month of February.

```
SQL> select mid,title,releasedate from movies where to_char(releasedate,'mm')=02;

       MID TITLE                                    RELEASEDA
---------- ---------------------------------------- ---------
       105 Daddy Cool                               14-FEB-09
       107 Pulp Fiction                             14-FEB-20
       110 Oppenheimer                              21-FEB-23
```

6. Illustrate the different date functions using dual table (to_date, Add_months, last_day, months_between, next_day, round etc.)

```
SQL> select
  2  to_date(sysdate)TOD,add_months(sysdate,4)ADDD,last_day(sysdate)LASTD,months_between('19-mar^B2023','19-mar-2022')MONB,next_day(sysdate,'Sunday')NEXTD,round(sysdate,'YYYY')RUND from dua
l;

TOD       ADDD      LASTD          MONB NEXTD     RUND
--------- --------- --------- ---------- --------- ---------
04-SEP-23 04-JAN-24 30-SEP-23         12 10-SEP-23 01-JAN-24
```

```
SQL> select
  2  to_date(sysdate)TOD,add_months(sysdate,4)ADDD,last_day(sysdate)LASTD,months_between('19-mar
l;

TOD       ADDD      LASTD          MONB NEXTD     RUND
--------- --------- --------- ---------- --------- ---------
04-SEP-23 04-JAN-24 30-SEP-23         12 10-SEP-23 01-JAN-24
```

7. Illustration of special date formats using to_char function (use of th,sp,spth)

```
SQL> select TO_CHAR(SYSDATE, 'DDth'),TO_CHAR(SYSDATE, 'DDsp'),TO_CHAR(SYSDATE, 'DDspth') from dual;

TO_C TO_CHAR(SYSD TO_CHAR(SYSDAT
---- ----------- -------------
07TH SEVEN       SEVENTH
```

8. Calculate the total gross earnings for movies released after June 16, 2020.

```
SQL> select sum(gross) from movies where releasedate>'16-jun-2020';

SUM(GROSS)
----------
5849032179

SQL>
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.5 Set Operations

Date :

Query

1. Create a new table IndustryHit (Id, title, genre, Certificate, Gross, Releasedate). Insert some movies from Movies table and some new movies in the new table IndustryHit.

```
SQL> create table IndustryHit(id int primary key, title varchar2(100) not null, genre varchar
2(50), certificate varchar2(10), gross int, releasedate date);

Table created.

SQL>

SQL> INSERT INTO IndustryHit (Id, Title, Certificate, Gross, Releasedate) SELECT mid, Title,
Certificate, Gross, Releasedate FROM (SELECT mid, Title, Certificate, Gross, Releasedate FROM
 Movies ORDER BY ImdbRating DESC)WHERE ROWNUM <= 5;

5 rows created.

SQL> insert into industryhit values(200,'life of pi','adventure drama','U/A',6900000000.00,'2
8-sep-2012');

1 row created.

SQL> insert into industryhit values(202,'jailer','action thriller','U/A',5660000000.00,'10-au
g-2023');

1 row created.
```

2. Retrieve the titles of all movies and industry hits which are in the action thriller genre.

```
SQL> select title from movies union select title from industryhit where genre='action thrille
r';

TITLE
--------------------------------------------------------------------------------------------
-------
Avatar
Daddy Cool
Inception
Lucifer
Master
Mayaanadhi
Oppam
Oppenheimer
Pulp Fiction
The Avengers
Voice Of Sathyanathan

TITLE
--------------------------------------------------------------------------------------------
-------
jailer

12 rows selected.
```

3. Retrieve the titles of all movies including industry hits.

```
SQL> select title from movies where hit=1 union select title from industryhit;

TITLE
--------------------------------------------------------------------------------------------------------
-------
Avatar
Daddy Cool
Inception
Lucifer
Mayaanadhi
Oppam
Oppenheimer
Pulp Fiction
The Avengers
Voice Of Sathyanathan
jailer

TITLE
--------------------------------------------------------------------------------------------------------
-------
life of pi

12 rows selected.
```

4. Retrieve the titles of all movies which are not industry hits.

```
SQL> SELECT Title FROM Movies MINUS SELECT Title FROM Industryhit;

TITLE
--------------------------------------------------------------------------------------------------------
-------
Avatar
Daddy Cool
Master
Mayaanadhi
Oppam
Voice Of Sathyanathan

6 rows selected.
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.6 Illustration of Group By having clause

Date :

Query

1. For all genres, display genre type and the sum of all Gross for each genre. Name the derived column SUM_COLL.

```
SQL> select genre,sum(gross) as sum_coll from industryhit group by genre;

GENRE                                              SUM_COLL
-------------------------------------------------- ----------
                                                    523593411
adventure drama                                    6900000000
action thriller                                    5660000000
```

2. For all genres, display the genre type and the number of titles. Name the derived column TITLE_COUNT.

```
SQL> SELECT Genre, count(Title) AS Count FROM Industryhit GROUP BY Genre;

GENRE                                                COUNT
-------------------------------------------------- ----------
                                                          5
adventure drama                                           1
action thriller                                           1
```

3. Display the genres which have more than 3 titles.

```
SQL>  select genre,count(title) from industryhit group by genre having count(title)>3;

GENRE                                              COUNT(TITLE)
-------------------------------------------------- ------------
                                                              5
```

4. Retrieve the total number of movies released in each year, only for years with at least 5 movies.

```
SQL> select count(title) from movies group by releasedate having count(title)>5;

no rows selected
```

5. List the certificates along with the number of movies for each certificate, but only show certificates with more than 3 movies.

```
SQL> select certificate,count(title) from movies group by certificate having count(title)>3;

CERTIFICATE          COUNT(TITLE)
-------------------- ------------
PG-13                           4
```

6. Show the total gross earnings for each certificate, but only for certificates with total gross greater than $1 million.

```
SQL> select certificate,sum(gross) from movies group by certificate having sum(gross)>1000000
;

CERTIFICATE          SUM(GROSS)
-------------------- ----------
U/A                  5234500678
U                     761030000
R                     510142715
PG-13                  13693985
```

7. List the release years with the highest number of movies and the corresponding movie count, limited to the top 3 years.

```
SQL> select * FROM (select to_char(releasedate,'YYYY') YEAR,count(*) Count FROM Movies GROUP BY to_char(releasedate,'YYYY') ORDER BY Count DESC) where ROWNUM<=3;

YEAR      COUNT
---- ----------
2020          5
2023          3
2009          1
```

Activity # 3
Practice SQL Data Manipulation Language (DML) commands

Description : 3.7 Sub queries

Date :

Query

1. Retrieve the titles and runtime of movies with the highest Metascore.

```
SQL> SELECT Title, Runtime FROM Movies WHERE Metascore = (SELECT MAX(Metascore) FROM Movies);


TITLE                                       RUNTIME
---------------------------------------- ----------
Pulp Fiction                                  2.57
```

2. List the titles of movies with a Gross amount greater than the average Gross amount of all movies.

```
SQL> select title from movies where gross >(select avg(gross) from movies);

TITLE
----------------------------------------
Mayaanadhi
Daddy Cool
```

3. Retrieve the titles and descriptions of movies with a Metascore lower than the average Metascore.

```
SQL> select title,description from movies where metascore <(select avg(metascore) from movies
);

TITLE                                    DESCRIPTION
---------------------------------------- ----------------------------------------------------
--------------------------------------------------
Daddy Cool                               A quirky family comedy with a touch of mystery
Master                                   A suspenseful thriller that takes place under the co
ver of darkness.
Lucifer                                  A political thriller with unexpected twists

SQL>
```

4. List the movie titles and their IMDb ratings for movies released in the year with the highest average IMDb rating.

```
SQL> SELECT Title, ImdbRating, to_char(releasedate,'YYYY') Release_Year FROM Movies WHERE to_char(releasedate,'YYYY') = (select year from (SELECT to_char(releasedate,'YYYY') year,round(AVG(
har(releasedate,'YYYY') order by Avg(imdbrating) desc)where rownum=1);

TITLE                                    IMDBRATING RELE
---------------------------------------- ---------- ----
The Avengers                                      8 2012
```

5. Retrieve the movie titles and their IMDb ratings for movies that have a Metascore greater than twice their IMDb rating.

```
SQL> SELECT title, imdbRating FROM movies WHERE metascore > ALL (SELECT
  2  POWER(imdbRating, 2) FROM movies);

TITLE                                    IMDBRATING
---------------------------------------- ----------
Oppenheimer                                     8.1
Avatar                                          5.9
Oppam                                           7.5
Pulp Fiction                                    8.9
```

6. Find the title and gross amount of the top 3 highest-grossing movies.

```
SQL> SELECT Title, Gross FROM (select * from movies order by gross desc ) where
  2  rownum <=3;

TITLE                                        GROSS
---------------------------------------- ----------
Mayaanadhi                               5234500678
Daddy Cool                                650030000
Oppenheimer                               505000000
```

7. Calculate the total number of votes received by movies released in the year 2022.

```
SQL> SELECT SUM(Votes) AS Total_Votes FROM (SELECT * FROM MOVIES
  2  WHERE EXTRACT(Year from Releasedate)=2022);

TOTAL_VOTES
-----------
         94
```

8. List the titles and certificate ratings of movies that have an IMDb rating below the average IMDb rating.

```
SQL> SELECT Title, Certificate FROM movies WHERE ImdbRating < (SELECT
  2  AVG(ImdbRating) FROM Movies);

TITLE                                    CERTIFICATE
---------------------------------------- --------------------
Voice Of Sathyanathan                    U
Mayaanadhi                               U/A
Avatar                                   PG-13
Daddy Cool                               U
Master                                   R
```

LEARN • SERVE • EXCEL
RAJAGIRI

LEARN • SERVE • EXCEL
RAJAGIRI